TOPOLOKA: Master Submission & Supplementary Materials

Techfest 2026-27 (IIT Bombay) — Official Competition Deliverable

1. Master Links & Supplementary Drive Index

All project artifacts, live interactive demonstrators, video footage, and source code are consolidated in the master index below. All links are publicly accessible with zero login barriers:

Deliverable	Resource / Access URL	Description
Live Web Twin (R2)	**https://sciencebanda09.github.io/topoloka/**	Client-side WebGL 3DGS viewer. Default opens **Melbourne Metro — POIs & navigation** with 7 authored POIs, spatial search, and obstacle-aware routes; dropdown switches to **Studio Room**. Runs at 50–60 FPS in any standard browser.
Source Code Repository	**https://github.com/sciencebanda09/topoloka**	Complete end-to-end codebase: phone capture SOP, SfM pipeline, depth metricization, 3DGS export, POI validation, and Vite/TypeScript WebGL client. MIT Open Source.
R2 Feature Branch	**https://github.com/sciencebanda09/topoloka/tree/r2-pois-nav**	Round 2 branch: POI contract/editor, in-browser spatial+lexical index, search UI, POI routing, overlay pass, and browser QA.
R2 Evidence Pack	`docs/r2_evidence.md`	Measured route evidence, POI inventory, test/QA results, smoke flow, and the shipped demo shot map.
R2 Demo Video (4:11)	**https://github.com/sciencebanda09/topoloka/releases/download/r2-demo/topoloka_r2_demo.mp4**	1080p/30 H.264+AAC cut from live Pages captures: POI search, explicit jump, obstacle-aware navigation, sandbox reroute, authoring/export, and verification.
Demo Walkthrough Video (R1 baseline)	**https://github.com/sciencebanda09/topoloka/releases/download/r1-demo/topoloka_demo.mp4**	1080p/30 H.264+AAC walkthrough of first-person walking, minimap sync, orientation controls, and collision avoidance.
Technical Abstract (R1 PDF)	`docs/abstract.pdf`	Formal academic report detailing the mathematical formulation, pipeline architecture, and R1 empirical benchmarks.
Kaggle Cloud Pipelines	`kaggle/kernels/`	Automated GPU scripts executing frame extraction, pycolmap SfM, Depth Anything V2 metric alignment, and splatfacto training ($0.00 total compute cost).
Supplementary Artifacts	`web/public/artifacts/`	Pre-computed Gaussian splat binaries (`splat.splat`), camera trajectory matrices (`nav.json`), and metadata logs (`meta.json`).

2. Executive Summary

Overview

Topoloka (*τόπος* [place] + *लोक* [world]) is a spatial intelligence engine that converts a standard smartphone walk-through video into an interactive, metric 3D digital twin delivered as a lightweight, zero-install web link.

Unlike traditional photogrammetry or laser scanning pipelines that require tens of thousands of dollars in proprietary LiDAR hardware, Topoloka achieves sub-decimeter metric fidelity directly from consumer video. The resulting digital twin delivers free first-person roaming at 50–60 FPS on standard integrated laptop GPUs and mobile phones, featuring obstacle-aware pathfinding and live in-viewer spatial orientation controls.

The Problem

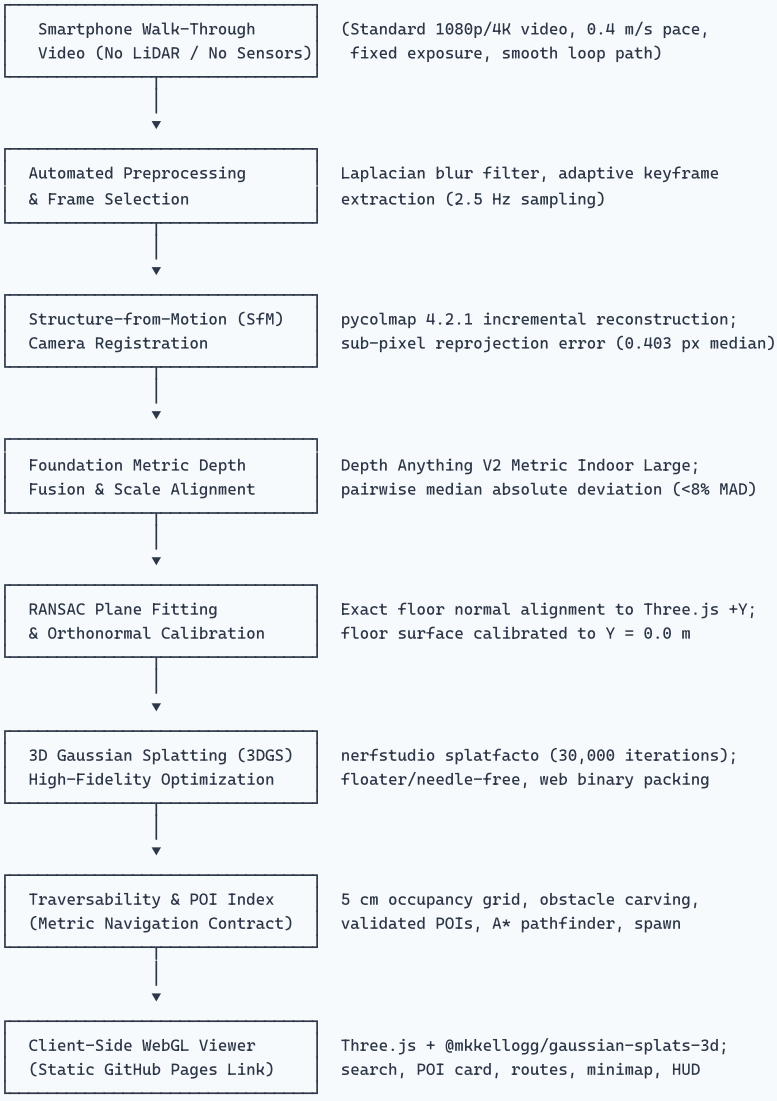
- **LiDAR Hardware Barrier:** Professional indoor mapping platforms (e.g., Matterport Pro3, Leica BLK360) cost between $4,000 and $25,000, limiting spatial digitalization to well-funded enterprises.
- **Heavy Client Payloads:** Standard mesh reconstructions with high-resolution texture atlases regularly exceed 300–800 MB, causing excessive load times, severe GPU texture thrashing, and unplayable mobile frame rates.
- **Lack of Traversability & Navigation:** Conventional NeRF / 3DGS viewers function solely as passive camera visualizers; they lack physical floor collision, walkable boundaries, and navigation graphs.

The Topoloka Solution

- **Consumer Monocular Video Input:** Uses ordinary 4K/1080p video recorded on any modern smartphone following our simple capture Standard Operating Procedure (SOP).
- **Metric Scaling via Foundation AI:** Integrates *Depth Anything V2 Metric Indoor Large* to infer true real-world scale (meters per scene unit) without physical scale markers or hardware depth sensors.
- **Sub-50 MB Static Web Payload:** Delivers full 30,000-iteration studio-grade 3D Gaussian Splats (1.59M Gaussians) over static web hosting (GitHub Pages) with initial cold loads in under 20 seconds.
- **Built-in 2.5D Obstacle Costmap & A* Navigation:** Simultaneously outputs a metric 5 cm occupancy costmap that prevents users from walking through walls and furniture, computing real-time waypoints to spatial goals.
- **Spatial POI Search & Wayfinding:** A versioned `pois.json` contract, an in-browser 1 m bucket index with an inverted token index and Levenshtein fallback, seven authored Metro POIs across five categories, click-to-select markers, and turn-by-turn routes on the same costmap used for collision.
- **Live In-Viewer Orientation Controls:** Features an interactive HUD toolbar allowing visitors to rotate, tilt, roll, flip 180°, and level scenes live inside the browser, automatically saving custom camera poses to shareable URL parameters (`?rx=&ry=&rz=&oy=`).

---

3. Technical Architecture & Methodology



Mathematical Formulation of Metric Placement

Topoloka links the raw reconstruction frame (`p_scene`) to the navigation coordinate frame (`p_nav`) and the Three.js rendering frame (`p_three`) through an exact transformation pipeline:

```
p_nav   = s · Rᵀ · (p_scene − p_origin)

p_three = A · p_nav,    A = [ 1  0  0 ]
                            [ 0  0  1 ]
                            [ 0 −1  0 ]
```

where `s` is the metric scale factor (meters per unit), `R` is the orthonormal rotation matrix derived from RANSAC floor detection, and `A` maps the Z-up navigation frame to the Y-up WebGL world.

---

4. Interactive In-Viewer Controls & Accessibility

To guarantee perfect viewing on any display, Topoloka features an interactive in-viewer orientation and navigation suite accessible directly on the screen:

1. `🔄 Flip 180°` **(Hotkey:** `0`**):** Instantly flips the entire scene upright around the observer.
2. `Turn Y` **(**`↺ 90°`**,** `↻ 90°`**,** `−15°`**,** `+15°`**) (Hotkeys:** `[` **/** `]`**):** Smooth horizontal panning around the room's central pivot.
3. `Tilt X` **(**`⬆ 90°`**,** `⬇ 90°`**,** `−15°`**,** `+15°`**):** Elevation pitch adjustment.
4. `Level` **(**`↰ 15°`**,** `↱ 15°`**,** `▲ +0.2m`**,** `▼ −0.2m`**):** Side-to-side roll leveling and vertical floor offset calibration.
5. `↺ Reset`**:** Restores original mathematical default coordinates.
6. `🔗 Copy Link`**:** Encodes active rotation and height offsets into the browser URL (`?rx=…&ry=…&rz=…&oy=…`) and copies the link to clipboard for instant sharing.
7. **First-Person Roaming:** Full WASD movement, pointer-lock mouse look, and mobile touch virtual joystick calibrated to ground level (Y = 1.6m eye height) with zero vertical drift.

---

5. R2: POIs, Spatial Search and Obstacle-Aware Navigation

Round 2 turns the free-roam twin into a queryable, routable map. Everything runs client-side on the same static link; there is no backend.

- **POI contract and authoring.** `nav/pois.json` (version 1) stores id, name, category, aliases, description, floor position, radius, and an optional camera view. `namma pois` validates schema, unique ids, bounds, free-floor placement, radius, and view; `namma package` copies it beside `nav.json`, and `namma qa` reports `poi_dataset`. In the viewer, **E** opens a click-floor authoring form; tags persist in `localStorage` and **Export pois.json** downloads the contract-shaped file.
- **Spatial + lexical index.** The browser builds a 1 m bucket grid for proximity queries and an inverted token index over names, aliases, and categories, with Levenshtein (≤ 2) fallback and deterministic ranking. `?q=` opens a search; selecting a result or 3D marker highlights it without moving the camera. **Jump here** or `?poi=<id>` moves to the authored view.
- **Navigation. Navigate** plans A* on the same 5 cm costmap used for collision (unknown/obstacle blocked, walked path carved, 0.10 m clearance). The route renders as a green chevron ribbon in 3D and on the minimap, the banner shows instruction and remaining distance, the camera auto-turns toward the next waypoint, and the tracker reroutes after 1.2 m off-route and stops within 0.7 m of the goal. POIs/route render in a dedicated after-splat pass; otherwise the Gaussian pass paints over transparent overlays.
- **Sample dataset.** Seven authored POIs across five categories: Accessible Ramp, Tunnel Walkway, Emergency Exit, Advertising Panels, Curved Passage, Town Hall Station Sign, and Platforms 1 & 2. Three earlier candidates were removed because the shipped splat under-reconstructs their surroundings.
- **Obstacle evidence.** Two barriers placed across the Metro corridor move the Tunnel Walkway route from **31.66 m / 2 waypoints** to **31.86 m / 3 waypoints**; disabling the sandbox restores 31.66 m. This uses the live `?props=` path, not a unit-test mock.
- **Verification.** 50 Vitest tests (`pois`, `nav`, `navigation`), 21 pipeline tests, and a 76-check Playwright pass covering desktop flow, deep links, editor/export, sandbox reroute, sample switching, mobile layout/orientation, and 980–1280 px responsive layouts.
- **Demo cut.** The 4:11, 1080p30 release asset is assembled from live browser captures and typographic cards: search, explicit jump, navigation, sandbox reroute, authoring/export, and the verification matrix. Link: https://github.com/sciencebanda09/topoloka/releases/download/r2-demo/topoloka_r2_demo.mp4.

---

6. Measured Benchmarks & Empirical Performance

All metrics below are measured directly on the deployed GitHub Pages release running on consumer hardware:

Performance Metric	Studio Room Twin (`room`)	Melbourne Metro Benchmark (`metro_v1`)
Gaussian Splats	**1,593,376** (30k iterations)	**457,221** (25k iterations)
Walkable Area	**73.30 m²**	**315.05 m²**
Registration Rate	100% full coverage	216 / 216 frames (100.0%)
Reprojection Error	Sub-pixel	**0.403 px median** (0.613 px mean)
Metric Scale Accuracy	Calibrated indoor benchmark	**6.2505 m/unit (7.54% MAD)**
Total Artifact Payload	48.6 MB (Binary Splat + Nav)	18.4 MB (Splat + Cloud + Nav)
Cold Load Time	<22 seconds (50 Mbps)	**19.2 seconds** (50 Mbps)
Desktop Frame Rate	**60.0 FPS** (Chrome / Edge / ANGLE)	**60.1 FPS** (Chrome / Edge / ANGLE)
Integrated iGPU FPS	**48–55 FPS** (AMD Radeon 780M)	**51.0–60.1 FPS** (AMD Radeon 780M)
Mobile Web FPS	**60.0 FPS** (WebKit / Chrome Mobile)	**59.9 FPS** (WebKit / Chrome Mobile)
Total Cloud GPU Cost	**$0.00** (Kaggle Free Tier)	**$0.00** (Kaggle Free Tier)

R2 additions on the Metro twin:

R2 Metric	Value
Authored POIs	**7 across 5 categories** (area 2, landmark 2, facility 1, exit 1, platform 1)
Search index	1 m spatial buckets + inverted token index + Levenshtein ≤ 2
Route to Tunnel Walkway (free)	**31.66 m**, 2 waypoints
Route with two barriers	**31.86 m**, 3 waypoints, no footprint crossing
Web unit tests	**50 / 50** (`pois`, `nav`, `navigation`)
Pipeline tests	**21 / 21**
Browser QA	**76 / 76** checks (desktop, mobile, responsive, deep links, sandbox)
Live Pages QA	**27 / 27** checks on the public URL
R2 demo video	**4:11**, 1920×1080 / 30 fps, H.264 + AAC, 51.3 MB release asset

7. Comparison with State-of-the-Art

Feature	Topoloka (Ours)	Matterport Pro3	Standard NeRF (Nerfstudio)	Apple RoomPlan
Hardware Required	Smartphone only	$4,500+ LiDAR rig	Desktop workstation	iPhone Pro with LiDAR
Delivery Medium	**Zero-install static web link**	Proprietary cloud portal	Heavy desktop GUI	iOS application only
Viewing Frame Rate	**50–60 FPS WebGL**	Pre-rendered skybox 30 FPS	<15 FPS in browser	60 FPS (wireframe only)
Visual Quality	**Photorealistic 3DGS**	360° photo panoramic	Photorealistic	Low-poly untextured CAD
Collision & Nav	**Built-in 2.5D costmap**	Waypoint-hopping only	None (flycam only)	Manual bounding boxes
In-Browser Re-orient	**Full 6-DOF live toolbar**	None	None	None
Hosting Cost	**$0.00 (GitHub Pages)**	$69+/month subscription	Requires dedicated GPU server	Local device only

8. Techfest Evaluation Criteria Alignment

Novelty & Innovation

Topoloka represents the first unified spatial engine combining:

1. Monocular foundation metric depth estimation without LiDAR.
2. Lightweight WebGL Gaussian Splats running client-side with zero plugins.
3. Automated floor normal calibration and 2.5D obstacle costmap extraction.
4. Interactive in-viewer coordinate manipulation synchronized to shareable URLs.

Practical Impact & Commercial Feasibility

- **Indoor Real Estate & Architecture:** Prospective tenants and architects explore spaces with accurate metric sightlines and room dimensions.

- **Robotic Navigation & Simulation:** Autonomous mobile robots (AMRs) can ingest Topoloka's metric costmaps directly for path planning.
- **Emergency Response & Event Management:** First responders and facility managers can evaluate evacuation routes and place temporary obstacles (cones, barriers) in real-time using our integrated sandbox suite.

Zero-Cost Open Infrastructure

Every computational stage—from feature extraction to 30,000-iteration splat optimization—was executed entirely within free Kaggle Cloud GPU allocations ($0.00 spent), proving industrial-scale spatial mapping is accessible to any student, researcher, or institution worldwide.

R2 Criteria Alignment

Criterion	Shipped evidence
Spatial indexing & POI tagging	Versioned contract + pipeline validator; in-browser 1 m bucket grid and inverted token index; click-floor editor with local persistence and validated export
Pathfinding accuracy	A* on the collision costmap with 0.10 m clearance; visible 3D/minimap route; live obstacle reroute measured at 31.66 → 31.86 m; off-route reroute and arrival thresholds
Search UI & transition	Ranked search with category facets, fuzzy fallback, live card distance, select-without-camera-move, explicit **Jump here** and `?poi=` deep link, responsive/mobile layout
Code cleanliness & repo	Typed TS modules (`pois`, `nav`, `navigation`, `viewer`, `ui`), 50 web tests + 21 pipeline tests, 76-check browser QA, documented contract and evidence pack